Finance Notes - List of Useful formulae for FM

Summary Formulae for FM

CHAPTER - RATIOS

Liquidity Ratios

It helps in analysing the ability of the company to meet its **short-term** obligations towards the various stakeholders.

Ratio	Unit	Formula	Remarks
Working capital	In Rs.	Current assets – current liabilities	Should be positive to depict a comfortable position
Current Ratio	times	Current assets / current liabilities	Should be more than 1 >1.33 and < 2 is considered to be a comfortable position for any company
Quick Ratio/Acid Test Ratio	times	(Current assets – inventory, other prepaid expenses) / current liabilities **OR** (Cash/Cash equivalents + marketable securities + accounts receivable) / current liabilities	It shows the proportion of assets that are quickly convertible into cash; it should therefore exclude inventory and any other prepaid expenses/advance tax from the current assets.

Solvency Ratios

It helps in analysing the ability of the company to meet its **long-term** contractual obligations towards the various stakeholders. It depicts how well the company is capitalised and if there can be any danger to its long term existence.

Ratio	Unit	Formula	Remarks
Net-worth	In Rs.	Equity + reserves & surplus – revaluation reserve	Also known as shareholder equity/owners' equity. If there is any revaluation reserve then it has to be deducted to calculate net-worth/shareholder equity.
Capital Employed	In Rs.	Net-worth + Long term Debt + other long term liabilities **OR** Total Assets – Current Liabilities	Sources of funds employed in the company
Debt Equity Ratio/ Leverage Ratio	Times	Long term debt/Net-worth	It shows the ratio of outsider funds as compared to own funds. Only long term debt of company is taken
Overall Gearing	Times	Total debt/Net-worth	It shows total outsider funds, both long term and short term borrowing, as compared to own equity in the firm.
Net-worth Ratio / Proprietary Ratio	Times	Net-worth/Total assets	Higher the ratio, better the LT solvency of the company
Debt to Capital Employed Ratio	Times	Long term Debt /Capital Employed	

Ratio	Unit	Formula	Remarks
Total Assets to Debt Ratio	Times	Total Assets / long term debt	
Financial Leverage or Equity Multiplier	Times	Average Total Assets / average total equity	Useful in DU-Pont analysis
Debt Service coverage Ratio	times	Net Operating Income/Total debt service cost	Net Operating Income = Net Profit after Tax + Depreciation/Amortization + Interest + other non-cash items Total debt service cost = principal repayment + interest payment + any other sinking fund obligation
Interest coverage Ratio	times	Operating Income/Interest payment	Operating Income = EBIT

Note: Operating Income and net operating income differ on account of tax

Turnover/ Activity/ Efficiency Ratios

It helps in analysing how efficiently the company utilises its resources to operate efficiently i.e. how efficiently it converts its assets into profits.

Ratio	Unit	Formula	Remarks
Account Receivables Turnover	times	Credit Sales/ average receivables	Annual sales from Income statement for a period is taken so to compare it with a Balance sheet figure, we should take the average of receivables from the beginning of period and end of period Balance Sheet. It shows the speed with which credit sales are realised
Days of sales outstanding	Days	365/receivables turnover	Shows the number of days in which sales is paid; Lesser the better.
Inventory Turnover	times	COGS/ average inventory	No. of times inventory is converted into revenue
Days of inventory in hand	Days	365/inventory turnover	No. of days of inventory in hand of company
Payable Turnover	times	Purchases/ average payables	
No. of Days of payables	Days	365/payables turnover	
Total assets turnover	times	sales/ average total assets	Indicates the extent to which total assets results in sales
Net assets (or capital employed) turnover	times	Revenue from operations/ average capital employed	
Fixed assets turnover	times	Revenue from operations/ average fixed assets	

Ratio	Unit	Formula	Remarks
Working Capital turnover	times	Revenue from operations/ average working capital	
Working capital cycle or operating cycle	days	Receivable days + inventory days – payable days	

Profitability Ratios

It depicts the profitability of the company in relation to its revenue and assets employed in the business. It helps to understand the earning capacity of the company.

Ratio	Unit	Formula	Remarks
Gross Profit	In Rs.	Net Sales - COGS	
Gross Profit Margin	In %	Gross Profit/Net Revenue * 100	
Operating Profit or EBIT	In Rs.	Gross Profit – Operating expenses - Depreciation	
Operating Profit Margin	In %	Operating Profit/Net Revenue * 100	
Net Profit Ratio	In %	Net Profit/Net revenue *100	
Return on Assets (ROA)	In %	Net income/average total assets *100	
Return on Equity (ROE)	In %	Net income / average equity * 100	
Return on Capital Employed (ROCE)	In %	EBIT / average capital employed *100	
Du-Pont Analysis	In %	RoE = Net profit Ratio * Asset turnover ratio * Equity Multiplier	It is a breakdown of profitability to see what has contributed to the RoE of the companys
Shareholder Ratios			
Earnings per Share (EPS)	Rs.	Profit available to equity shareholders/ no. of equity shares	
Price Earnings Ratio (P/E Ratio)	Times	Market price per share/ EPS	
Dividend Payout Ratio	%	Dividend/earnings *100	
Dividend per share	Rs.	Dividend paid/no. of shares	
Book Value per share	Rs.	Net-worth/no. of shares	
Per share based calculation			
Market Cap	Rs.	Market price * no. of shares	
Networth	Rs.	Book value per share * no. of shares	
Issued equity capital	Rs.	Face value * no. of shares	

CHAPTER – CAPM

Ratio	Unit	Formula	Remarks
Cost of Equity Capital (CAPM Model) (used for expected return on a portfolio)	%	Risk free rate + Beta (Market rate – risk free rate)	Risk free rate can be represented by yield of 10 year Govt bond (Market rate – risk free rate) is also known as risk premium
Cost of Debt	%	Interest rate * (1-tax)	After tax cost is taken as interest on debt is a tax deductible item
WACC (weighted average cost of capital)	%	WACC = (Cost of equity * proportion of equity in total capital) + (Cost of debt * proportion of debt in total capital)	Total capital = equity + LT debt = E + D Proportion of equity = E/ (E+D) Proportion of Debt = D/ (E+ D)
Cost of equity (Dividend discount Model)	%	(Expected Dividend / CMP) + growth rate	Expected Dividend means the dividend to be paid next year or end of the year CMP = current market price of the equity share

CHAPTER – BREAK-EVEN ANALYSIS & OTHER BASIC FINANCE

Ratio	Unit	Formula	Remarks
Break even quantity	No.	Fixed cost/(Sales – Variable cost)	(Sales – Variable cost) is referred to as the contribution
PV Ratio	times	Contribution/sales	PV stands for Profit volume ratio Here per unit contribution and sales is taken
Break Even sales	In Rs.	Fixed cost/PV ratio	This gives the break even sales in rupee terms and not the no. of units sold
Margin of safety	%	Total sales – break-even sales	
Operating leverage	times	Contribution/EBIT	Contribution = sales – variable cost The ratio denotes how the fixed cost can help increase the operating profitability of the company
Financial leverage	times	EBIT/PBT	The ratio denotes how the fixed cost debt (i.e. using leverage or debt funding) can help increase the profitability of the company
Combined leverage	times	Operating leverage * financial leverage **OR**	

Ratio	Unit	Formula	Remarks
		Contribution/PBT	

Finance Notes - List of Useful formulae for FM

CHAPTER – BONDS

Ratio	Unit	Formula	Remarks
Bond Price of Bond Value (coupon paying)	Rs.	*Bond price=* $C * \frac{\left[1-\left[\frac{1}{(1+i)^n}\right]\right]}{i} + \frac{RV}{(1+i)^n}$	C = coupon i = interest rate/YTM RV = redemption value or maturity value n = maturity period/no. of payment period
Zero coupon Bond price	Rs.	*Bond price=* $\frac{RV}{(1+i)^n}$	Since no coupon, only the redemption value is discounted to present value
Perpetuity Bond price	Rs.	C/i	Since no maturity is there, n and RV does not exist C = coupon i = interest rate/YTM
Current Yield	%	Coupon/current market price of bond	
YTM (approx)	%	*YTM = (C + (FV-MP)/n)/ (FV+MP)/2*	C = coupon FV = face value MP = market price N = maturity period
Real rate of interest/effective interest rate	%	1 + R(n) = (1+r) * (1 + inflation rate)	R(n) = nominal rate of interest r = real rate of interest
Duration	years	*D = (SUM of (PV of inflow* no of years to inflow))/value of bond*	Known as Macaulay duration
Modified duration	years	d = D/ (1 + i/m)	d = modified duration D = macaulay duration i = interest rate m = number of coupon payments in a year (compounding in a year)
Duration (as a measure of interest rate risk)	years	D = change in price of bond/change in interest rate	
Duration of zero coupon bond	years	D = maturity of zero coupon bond	
Duration of perpetuity bond	years	D = (1 + YTM)/YTM	
Dirty price of bond	Rs.	Clean price + accrued interest	

CHAPTER – DERIVATIVES

Ratio	Unit	Formula	Remarks
Option premium	Rs.	Intrinsic value + time value	Option premium is calculated using Black Scholes model which is not required to be known in detail. Option premium has 2 components – intrinsic value and time value
Intrinsic value of an option	Rs.	'In the moneyness' value of option If the option is 'out of money' then it is zero	
Put-call parity		C + PV of Strike price = P + spot price of underlying asset	Only applicable to European style options C = call option premium PV of strike = strike price/(1+r)^n P = put option premium Spot price = current market price

CHAPTER – CAPITAL BUDGETING

Ratio	Unit	Formula	Remarks
NPV (Net present value)	Rs.	PV of inflows – PV of Outflows Where PV is calculated as below: $NPV = -C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + ... + \frac{C_T}{(1+r)^T}$ $-C_0 = Initial\ Investment$ $C = Cash\ Flow$ $r = Discount\ Rate$ $T = Time$	Positive NPV or NPV>0 is considered to be good to accept a project
IRR (internal rate of return)	%	It is done through hit and trial method to calculate r in the NPV formula above	The rate at which NPV is zero
PI (profitability index)	times	PV of Inflows/PV of outflows	Similar to NPV in the sense that NPV is in absolute rupee terms while PI is in ratio format As such when NPV = 0, PI will be 1 PI> 1 to accept project
Payback period	years	***In case of same inflow each year:*** Payback period = Initial investment/annual cash inflow ***In case of uneven inflow each year:*** Payback period = Initial investment - annual cash inflow of each year till it becomes 0.	It is the number of years it takes for the inflow to recover the cost

Note: NPV, IRR and PI are time value adjusted ways of capital budgeting while payback period ignores time value of money; a discounted payback period is also sometime calculated in which instead of taking the absolute amount of inflows, the discounted inflows are considered.

CHAPTER – CERTAIN BANK SPECIFIC RATIOS

Ratio	Unit	Formula	Remarks
NII (Net interest income)	Rs.	Interest earned – interest expense	It is like saying the gross income for a bank
Interest yield	%	Interest earned/average interest earning assets	Interest earning assets are mostly just advances given; However if there are some investments like debt mutual fund on which interest is earned then that is also taken as interest earning assets
Cost of deposits	%	Interest paid on deposits/average deposits	
Cost of funds	%	Total interest paid/ average interest bearing liabilities	Cost of deposits is a part of cost of funds. When a bank also has various borrowings or bond issues as liabilities then interest bearing liabilities will include deposits plus other borrowings
Interest spread	%	Interest yield – cost of funds	
NIM (net interest margin)	%	NII/average interest earning assets	
Cost to income	%	Non interest expenditure / Net Total Income * 100	Non-interest expense is majorly operating expenses Net total income = NII + other income OR Total income – interest paid
CAR (Capital Adequacy Ratio) OR CRAR (capital to risk (weighted) assets ratio)	%	Total capital/ risk weighted assets	Total capital = Tier I capital + Tier II capital